

Week 4 Project Summary

Track: Machine Learning Engineering - HealthConnect No-Show Prediction

1. Track-Specific Problem

Design the initial structure of a reliable, reproducible machine learning system that predicts the likelihood of a scheduled HealthConnect appointment ending in a no-show, early enough for clinic staff to intervene.

2. Resources Used

- HealthConnect_Appointment_Data.csv — 5,000 appointment records, 18 columns
- HealthConnect_Data_Dictionary.xlsx — field definitions and notes for all 18 columns

3. Key Observations from Week 4

- appointment_outcome has three classes (Attended 46.3%, No-Show 48.5%, Cancelled 5.3%); Cancelled is scoped out of the v1 binary model as a distinct, already-known-in-advance event.
- waiting_time_minutes is populated for the large majority of No-Show and Cancelled rows, which is logically inconsistent with a “time spent waiting” metric flagged as a likely data leakage risk and excluded from candidate inputs pending clarification.
- reminder_channel's 1,366 missing values align exactly with reminder_sent = No , a logical pattern, not a data defect.
- distance_to_clinic_km and waiting_time_minutes have small amounts of missingness (1.8% and 1.2%) spread evenly across outcome classes genuine gaps needing an imputation decision.
- previous_no_shows already shows a strong, clean signal in the raw data: no-show rate rises from 43.5% (no prior no-shows) to 68.8% (3+ prior no-shows).
- All internal consistency checks passed: booking_lead_days, appointment_day, and age_group are all correctly derivable from their source columns, and previous_no_shows never exceeds previous_appointments.

4. Proposed Approach

A binary classification system (No-Show vs Attended) scored in batch ahead of each appointment, using pre-appointment-only features (demographics, appointment details, patient history, reminder information, distance), with a temporal train/test split to reflect real production conditions, versioned model artifacts, and integration into the clinic's reminder/ops workflow. Full detail in the accompanying ML System Design Document.

5. Key Considerations Affecting the Project

- Data leakage risk around waiting_time_minutes needs resolution with the data team before feature finalization.
- Fairness review required if age/gender are used as direct inputs, given the operational actions (overbooking, reminder prioritization) the score could influence.
- Coordination needed with the Data Analytics and Data Science tracks working on the same dataset, to keep target-variable and feature scoping decisions aligned across tracks.
- Dataset is synthetic , real deployment would need re-validation against actual clinic behavior and ongoing drift monitoring.

6. Proposed Focus for Week 5

Build the reproducible data ingestion, validation, and feature-engineering pipeline; implement the temporal train/test split; resolve the `waiting_time_minutes` and missing-value questions; and produce a first documented baseline model.